

Mayank Bansal

AI/ML Engineering Student

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PROJECTS

PDF Question Answering System (RAG) | FAISS, Sentence Transformers, Flask, Gemini

- Built a **Retrieval-Augmented Generation (RAG)** system that answers questions strictly from PDF documents using semantic retrieval and guarded generation.
- Implemented document chunking with overlap, vector embeddings using Sentence Transformers, and similarity search with FAISS to retrieve relevant context.
- Designed prompt-level guardrails to prevent hallucinations and return explicit “I don’t know” responses for out-of-scope queries.
- Developed a Flask backend with a clean HTML, CSS, and JavaScript frontend, and added source attribution to make responses transparent and verifiable.

WasteSnap – Real-Time Waste Classification | *MobileNetV2, Flask, JavaScript, Render*

- Developed and deployed an AI-powered application to classify 9 types of waste using a fine-tuned MobileNetV2 architecture, achieving ~61% validation accuracy on a custom dataset.
- Engineered a production-ready Flask backend with Gunicorn on Render to handle image uploads and live camera feeds.
- Implemented real-time recyclability tagging and disposal tips on the frontend to drive eco-friendly user behaviour.

DigiDetect – Handwritten Digit Recognition | *TensorFlow, Flask, CNN*

- Built a full-stack web app featuring a custom Convolutional Neural Network (CNN) that achieved 98.9% test accuracy on the MNIST dataset.
- Designed a dynamic canvas interface for users to draw digits, processing inputs via a Flask API to return real-time predictions and confidence scores.
- Implemented robust image preprocessing pipelines to ensure consistent model performance on user-generated inputs.

Mood2Mail – Real-Time Email Tone Analyzer | *TF-IDF, Naive Bayes, Flask*

- Created a real-time tone detection system utilizing TF-IDF vectorization and Naive Bayes to classify email drafts into 5 distinct tone categories.
- Developed a seamless user experience using JavaScript fetch APIs to provide live tone feedback as the user types.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, HTML5, CSS3
- **Machine Learning & NLP:** TensorFlow, Scikit-learn, Retrieval Augmented Generation (RAG), Sentence Transformers, FAISS, TF-IDF, NumPy, Pandas
- **Web Development:** Flask, REST APIs, Gunicorn, Render Deployment, UI Design
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook

CERTIFICATIONS

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| • Complete Data Science and Machine Learning Skill Up – GeeksForGeeks | Nov 2025 |
| • AI Tools Skill Up – GeeksForGeeks | Oct 2025 |
| • Elements of AI – University of Helsinki | April 2025 |

ACHIEVEMENTS

- Solved 580+ questions on GeeksForGeeks including all difficulties
- Maintained a daily streak of 101 days of problem solving on GFG
- Ranked 159 in institute out of 39000+ on GFG with a coding score of 1627

EDUCATION

LOVELY PROFESSIONAL UNIVERSITY (TGPA – 9.33 / 10)
Bachelor of Technology in **Computer Science** Engineering

Phagwara, Punjab, India
August 2024– Present

ST. JOSEPH’S INTERNATIONAL SCHOOL
XII-SENIOR SECONDARY
Science (Physics, Chemistry and Mathematics)

Hisar, Haryana, India
April 2023-April 2024